MH2500 Probability and Introduction to Statistics

Slides for Week 3-2

Topics: Review of Bayes' formula, Independent events
Discrete Random Variables, CDF

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Bayes' Rule: review

Let A and $B_1, B_2, \ldots, B_n$ be events where the B_i's are disjoint, $\bigcup_{i=1}^{n} B_i = \Omega$, and $P(B_i) > 0$ for all i. Then

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}.$$

Bayes' rule: more examples

Example:

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a person who is not accident prone.

If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

Solution:

- Let A denote the event that the policyholder will have an accident within a year of purchasing the policy.
- Let B denote the event that the policyholder is accident prone.

Then $P(A|B) = 0.4$, $P(B) = 0.3$, $P(A|B') = 0.2$, $P(B') = 0.7$.

Hence, the desired probability is given by

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

$$= (0.4)(0.3) + (0.2)(0.7) = 0.26.$$

Example cont.

Suppose that a new policyholder has an accident within a year of purchasing a policy.

What is the probability that he/she is accident prone?

Solution: The desired probability is

$$P(B|A) = \frac{P(BA)}{P(A)} = \frac{P(B)P(A|B)}{P(A)}$$

$$= \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}.$$

It is an application of Bayes' rule — we just obtained

$$P(A) = P(A|B)P(B) + P(A|B')P(B') = 0.26.$$

One more

At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect.

Suppose, however, that a new piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered.

If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?

Solution: Let G denote the event that the suspect is guilty and C the event that he possesses the characteristic of the criminal.

Then $P(G) = 0.6$, $P(G') = 0.4$, $P(C|G) = 1$, $P(C|G') = 0.2$.

Thus, by Bayes' rule, the desired probability is

$$P(G|C) = \frac{P(GC)}{P(C)} = \frac{P(C|G)P(G)}{P(C|G)P(G) + P(C|G')P(G')}$$

$$= \frac{(1)(0.6)}{1(0.6) + (0.2)(0.4)} \approx 0.882.$$

Independent Events

Definition

Two events, A and B, are said to be **independent** if

$$P(A \cap B) = P(A)P(B).$$

Immediately, if A and B are independent and $P(B) \neq 0$, then

$$P(A) = \frac{P(A \cap B)}{P(B)} = P(A|B).$$

Intuitively, two events A and B are independent if knowing if B occurs or not does not effect the probability that A occurs, and vice versa.

Example

A fair coin is tossed twice. Let A denote the event that the first toss is a head, B the event that the second toss is a head, and C the event that exactly one head is obtained. Clearly A and B are independent events.

(i) Show that A and C are independent.

(ii) Is $P(A \cap B \cap C)$ equal to $P(A)P(B)P(C)$?

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(i) Show that A and C are independent.

(ii) Is $P(A \cap B \cap C)$ equal to $P(A)P(B)P(C)$?

Solution: First list A and C as

$$A = \{HT, HH\}, \quad C = \{HT, TH\}.$$

Then $P(A) = P(C) = \frac{1}{2}$ and $A \cap C = \{HT\}$. This gives

$$P(A \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(C)$$

and hence, A and C are independent.

(ii) $B = \{TH, HH\}$, and $P(B) = \frac{1}{2}$, and

- B and C are independent,
- A and B are independent.

Thus, A, B, C are pairwise independent.

We also have $P(A)P(B)P(C) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.

As $A \cap B \cap C = \emptyset$, $P(A \cap B \cap C) = 0$. This gives

$$P(A \cap B \cap C) \neq P(A)P(B)P(C).$$

Thus, three events A, B, C are not mutually independent.

Definition

Events $A_1, A_2, \ldots, A_n$ are **mutually independent** if for any subcollection $A_{j_1}, A_{j_2}, \ldots, A_{j_m}$,

$$P(A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_m}) = P(A_{j_1})P(A_{j_2}) \cdots P(A_{j_m}).$$

- Events are mutually independent if each event is independent from the set of the others.
- $A_1, A_2, \ldots, A_n$ are **pairwise independent**, if for any $i \neq j$, A_i and A_j are independent.
- Events that are mutually independent must be pairwise independent.
- The example above shows that events may be pairwise independent but not mutually independent.

Throw two fair dice and let (i, j) be the outcome. Define the events

$$A = \{(i, j) : i = 1, 2, 3\}, \quad B = \{(i, j) : i = 3, 4, 5\},$$

$$C = \{(i, j) : i + j = 9\}.$$

$$C = \{(3, 6), (4, 5), (5, 4), (6, 3)\}$$

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$$C = \{(3,6),(4,5),(5,4),(6,3)\} \Rightarrow P(C) = \frac{4}{36} = \frac{1}{9}$$

We also have $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$. By $A \cap B \cap C = \{(3,6)\}$,

$$P(A \cap B \cap C) = \frac{1}{36} = P(A)P(B)P(C)$$

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However,

$$P(A \cap B) = \frac{1}{6} \neq \frac{1}{2} \times \frac{1}{2} = P(A)P(B)$$

and hence A, B, C are not mutually independent.

Example

An infinite sequence of independent trials is to be performed. Each trial results in a success with probability p and a failure with probability $1 - p$. What is the probability that

1. at least 1 success occurs in the first n trials;
2. exactly k successes occur in the first n trials.

Answer: Let A denote the event that at least 1 success occurs in the first n trials. Then A' is the event that no success occurs in the first n trials. Thus

$$P(A') = (1-p)^n \text{ and } P(A) = 1 - P(A') = 1 - (1-p)^n.$$

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$$P(A') = (1-p)^n \text{ and } P(A) = 1 - P(A') = 1 - (1-p)^n.$$

Let B denote the event that exactly k successes occur in the first n trials. Then $P(B) = \binom{n}{k} \cdot p^k (1-p)^{n-k}$.

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Answer: Let A_i denote the event that the first i trials all succeed. Then

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots .$$

Let C denote the event that all trials result in successes. Then

$$C = \bigcap_{i=1}^{\infty} A_i$$

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Proposition

Suppose that $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots$, then

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{n\to\infty} P(A_n).$$

Proof: Consider the complements of A_n's. We have

$$A_1' \subseteq A_2' \subseteq \cdots \subseteq A_n' \subseteq \cdots$$

and hence

$$P\left(\bigcup_{i=1}^{\infty} A_i'\right) = \lim_{n\to\infty} P(A_n') = \lim_{n\to\infty}(1 - P(A_n)) = 1 - \lim_{n\to\infty} P(A_n).$$

By

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i'\right)',$$

we have

$$P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1 - P\left(\bigcup_{i=1}^{\infty} A_i'\right),$$

which is equal to

$$1 - [1 - \lim_{n\to\infty} P(A_n)] = \lim_{n\to\infty} P(A_n).$$

Example (cont.)

An infinite sequence of independent trials is to be performed. Each trial results in a success with probability p and a failure with probability $1-p$. What is the probability that all trials result in successes?

Answer: Let A_i denote the event that the first i trials all succeed and C the event that all trials result in successes. Then

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots$$

and

$$C = \bigcap_{n=1}^{\infty} A_n.$$

$$P(C) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} P(A_n) = \lim_{n\to\infty} p^n = \begin{cases} 0, & p < 1; \\ 1, & p = 1. \end{cases}$$

Discrete Random Variables

Two dice are rolled, and the sample space is

$$\Omega = \{(1,1),(1,2),\ldots,(1,6),(2,1),\ldots,(2,6),\ldots,(6,1),\ldots,(6,6)\}.$$

Suppose we count (i) number on the first die, (ii) sum of the numbers on two dice, (iii) product of the numbers on two dice.

- Since the outcome in Ω is random, each of these numbers are random numbers.

A **random variable** is a function

$$X : \Omega \to \mathbb{R}.$$

A **discrete random variable** is a random variable that can only take a finite or at most countably infinite number of values.

Example

A coin is flipped three times, and the sample space is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\},$$

Let X be the total number of heads.

$$P\{X = 0\} = \frac{1}{8}, \qquad P\{X = 1\} = \frac{3}{8}$$

$$P\{X = 2\} = \frac{3}{8}, \qquad P\{X = 3\} = \frac{1}{8}$$

Definition

Given a discrete random variable X taking values $x_1, x_2, \ldots$, its **probability mass function** or the **frequency function** of a random variable X is the function p where

$$p(x_i) = P\{X = x_i\}.$$

- It must hold that $\sum_{i=1}^{\infty} p(x_i) = 1$.

Example: A coin is flipped three times. Let X be the total number of heads.

$$p(0) = P(X = 0) = \frac{1}{8} \qquad p(1) = P(X = 1) = \frac{3}{8}$$
$$p(2) = P(X = 2) = \frac{3}{8} \qquad p(3) = P(X = 3) = \frac{1}{8}$$

Example continued

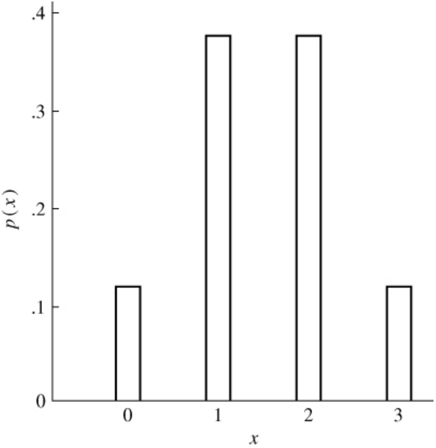

Figure 2.1 **A probability mass function.**

Definition

The **cumulative distribution function (cdf)** of a random variable is defined as the function $F : \mathbb{R} \to [0, 1]$ where

$$F(x) = P\{X \leq x\}, \qquad -\infty < x < \infty.$$

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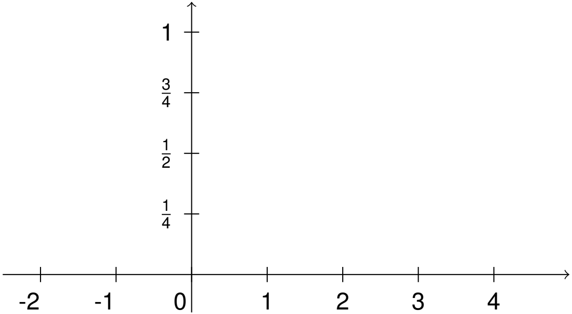
Convention:

We use uppercase for cumulative distribution function and lowercase letters for probability mass function.

Example: Graph the cdf of the previous example

$p(0) = \frac{1}{8}$, $p(1) = p(2) = \frac{3}{8}$, $p(3) = \frac{1}{8}$



. . . .

Note that F is always increasing, right-continuous and it always holds that

$$\lim_{x \to -\infty} F(x) = 0 \qquad \text{and} \qquad \lim_{x \to \infty} F(x) = 1.$$

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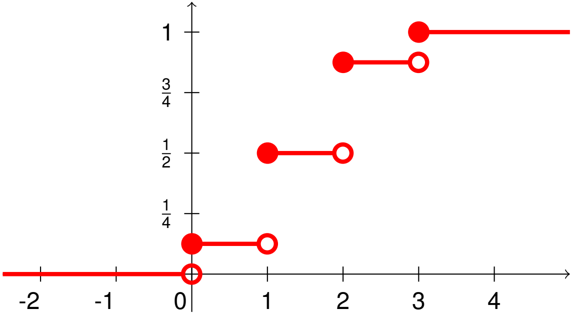

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$$\lim_{x\to-\infty} F(x) = 0 \qquad \text{and} \qquad \lim_{x\to\infty} F(x) = 1.$$

Use CDF to find

$$P\{a < X \leq b\}, \quad P\{X < a\}, \quad P\{a \leq X \leq b\}.$$

$$P\{a < X \leq b\} = P\{X \leq b\} - P\{X \leq a\} = F(b) - F(a)$$

$$P\{X < a\} = P\left(\bigcup_{n=1}^{\infty}\left\{X \leq a - \frac{1}{n}\right\}\right) = \lim_{n\to\infty} P\left\{X \leq a - \frac{1}{n}\right\}$$
$$= \lim_{n\to\infty} F\left(a - \frac{1}{n}\right) = F(a^-).$$

$$P\{a \leq X \leq b\} = P\{X \leq b\} - P\{X < a\} = F(b) - F(a^-)$$